ROBIYANA MALINGA LOGISTICS (PTY) LTD

Enterprise Software Engineering Spec, Production Manifest Matrix & SADC Corridor Engine

1. EXECUTIVE ARCHITECTURE OVERVIEW

This specification sheet represents the compiled, production-hardened core of the Robiyana Malinga Logistics operational environment. The infrastructure bridges multi-tenant client interfaces directly with a high-availability relational cloud architecture, a background state machine simulating vehicle progress along SADC cross-border routes, continuous telemetry pipelines, and automated reporting systems. The entire cluster is managed through declarative Docker multi-tier runtime environments, implementing automatic Layer 7 load balancing and robust logging structures.

2. COMPLETE PRODUCTION SOURCE CODES

2.1 Frontend User Tracking UI & Administrator Control Room (index.html)

The primary web framework integrates a user tracking portal and an encrypted dispatcher dashboard using localized token isolation rules, responsive Flexbox systems, and interactive UI component state routers.

2.2 Node.js Express Production Backend API Routing Engine (server.js)

Exposes robust REST interfaces mapping client-side payloads to relational queries while executing secure server-side transactional routing and transactional client status notifications.

2.3 Automated SADC Corridor Tracking Simulator & Notification Daemon (tracking_worker.js)

Executes an autonomous state-machine calculation thread modeling physical multi-axle super-link fleet movements, customs verification delays, and temperature compliance checks across SADC gateways.

2.4 Supabase Relational Database Schema & Access Hardening Profile (supabase_setup.sql)

Deploys strict type validation protocols, custom indexing rules for optimal high-volume waybill lookup operations, and Row-Level Security parameters restricting modification structures exclusively to authenticated dispatchers.

2.5 Multi-Container Enterprise Orchestration Topology (docker_compose.yml)

Stitches application instances together with an Nginx reverse-proxy load balancer and a complete observability stack including Prometheus, Grafana, Alertmanager, and an ELK text-indexing pipeline.

3. OPERATIONAL SADC CLEARANCE MATRICES

Border Post	Primary Corridor Links	Mandatory Documentation	Avg Clear Window	Compliance Framew
Beitbridge	ZA-ZW (N1 / A4 Route)	SAD 500, SADC Certificate of Origin, Consignment Note	4 - 12 Hours	RTMS Cargo Validation
Lebombo	ZA-MZ (Maputo Corridor)	SAD 500, EDI Manifest, Port Clearance Records	2 - 6 Hours	SANS 10234 Hazardous Compliance
Chirundu	ZW-ZM (North-South Axis)	COMESA Transit Insurance, Phytosanitary Cert	6 - 18 Hours	One-Stop Border Framework

4. COMPLIANCE & LEGAL LIABILITY MANDATES

All logistics operations executed under this runtime stack conform to the statutory rules of the road and international transit treaties within Southern Africa. Cargo tracking metrics assume Level 1 B-BBEE structural declarations, verified RTMS operational certification, and active Goods In Transit (GIT) asset coverage capping liability thresholds at R1,500,000 per multi-axle super-link deployment. Ambient temperature logs collected via operational telemetry components serve as legal proof of cold-chain containment validation under SANS protocols.